# The Iron Rush

*BUILD · Humanities · 40 minutes*

I can explain how ironstone, mines, rail, river and ironworks formed a change chain.

Name: ___________________________ Date: _______________

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

## We do · make a choice and give a reason

Commit before reveal: Did the find alone create the boom?

- No—mines, workers, transport and ironworks were also needed
- Yes—the rock changed itself
- No—ironstone had no use

My choice and the evidence I used:

## Resources and independent record

### BUILD Humanities · W11 · The Iron Rush

Enquiry: Why did ironstone found near Eston in 1850 change Teesside so quickly?

Learning objective: I can explain how ironstone, mines, rail, river and ironworks formed a change chain.

Inherited: Week 10 rail–river network and the 1825/1830 chronology.

CONSTRUCTED TEACHING SOURCE — created for classroom practice. It is not an archival original.

### Independent Humanities mission

A causal chain separating ironstone, pig iron, finished iron products and later steel, with at least one tested dependency.

Evidence target: A causal chain separating ironstone, pig iron, finished iron products and later steel, with at least one tested dependency.

### Supported

Order seven chain cards.

Evidence: Chain plus an oral or exactly scribed link.

Scaffold: First ___. Next ___. So ___.

### Standard

Add two because-arrows and label the 1850 trigger.

Evidence: Causal chain.

Scaffold: The find mattered because ___ and ___.

### Stretch

Remove one link and judge the impact.

Evidence: Counterfactual plus qualification.

Scaffold: Without ___, the rush would be slower because ___. However ___.

### Source/material record

Teacher-prepared constructed lab cards plus the locally approved source cards named for this lesson.

### Historian/geographer response & Influence

Pupil Humanities decision:

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Evidence or representation used:

Reasoning, limitation or uncertainty:

Audience genuinely received: describe through the agreed school workflow; no signature is required here.

Influence: The named class teacher uses the chosen Week 12 opener and rehearses raw → move → smelt → work → product first if needed.

Decision maker: Class teacher — replace with the adult's name before delivery.

Report back by: Start of Week 12 — add the calendar date before delivery.

## Learner source cards

ADAPTED SOURCE CARD

### HUM-B-W11-C01 · The 1850 Eston ironstone find

Provenance source: HUM-B-W11-S01

Tees Valley Museums records that a rich main seam of ironstone was found in the Eston Hills in 1850. Mines opened, railways carried ironstone to ironworks, and blast furnaces smelted it into iron. Steel production came later.

Can establish Supports the date, place and broad mine–rail–works sequence.

Cannot establish Prove that the discovery alone caused every change or describe every worker's experience.

Accessible route Four numbered facts: 1850 find; mines; rail movement; blast furnace makes iron. Ironstone, iron and later steel are different labels.

ADAPTED OFFICIAL-SYNTHESIS CARD

### HUM-B-W11-C02 · Industrial growth needs connected conditions

Provenance source: HUM-B-W11-S02

Historic England links Middlesbrough's rapid nineteenth-century growth to coal, the railway, the port, ironstone deposits and iron production. This supports a connected explanation rather than a one-cause story.

Can establish Supports broad chronology and the scale of connected industrial change.

Cannot establish Calculate the exact importance of each link or establish household and working conditions.

Accessible route Connection strip: raw material + transport + works + workers → rapid industrial growth.

## Source provenance

HUM-B-W11-S01 · Ironstone Miners Maker: Tees Valley Museums Group · Date: Undated live teaching resource Origin: Tees Valley Museums Group teaching resources Rights/use: © Tees Valley Museums Group. Facts adapted for classroom learning; no museum photograph is embedded because public republication rights are not stated. Access: Tactile or visual chain: ironstone → mine → rail → furnace → iron product.

HUM-B-W11-S02 · 11 and 13 Zetland Road — entry 1137392 Maker: Historic England · Date: Entry amended 5 March 2024 Origin: National Heritage List for England; Historic England Rights/use: Official list-entry text OGL v3.0. Credit © Historic England 2026, licensed under OGL v3.0. Embedded map and images excluded. Access: Large-date and proportion comparison card with plain text.

Print surface requires physical classroom checking before live use.

## Paper route • discuss the lesson choices

Use these choices with the matching teaching stage. Point to or number a choice, explain it, then revisit it after feedback. The original HTML keeps the live controls.

- blast furnace
- ironstone
- ironworking
- rail
- iron product
- pig iron
- Without the rail…
- Without the ironstone…
- Without the river and port…
- RAW ironstone
- MOVE mine
- MOVE rail wagon
- SMELT blast furnace
- SMELT pig iron
- WORK cast, or refine and roll
- PRODUCT iron rail or beam

| Choice / card | My reason or evidence | My revision after feedback |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

### Exit • one next step

After the adult has listened, what will you keep, improve or check next?